Shantanu Shirish Sirjoshi

Los Angeles, CA | 213 803 5287 | shantanusirjoshi@gmail.com | linkedin.com/in/shantanusirjoshi | github.com/dragonblade3k | dragonblade3k.github.io

MS Computer Science at USC, **expected May 2027**. 1.5 years building Java and Spring Boot settlement services for NSDL, India's national securities depository. Now builds LLM extraction pipelines on hosted and local models, with agentic coding tools as the primary development loop.

EDUCATION

University of Southern California · MS Computer Science · GPA 3.62 — **Expected May 2027**
Analysis of Algorithms, Machine Learning, Deep Learning, Information Retrieval, Foundations of AI, Agentic AI, Web Technologies

University of Mumbai · BE Computer Engineering · GPA 8.82 / 10 — May 2023

EXPERIENCE

Tata Consultancy Services · Systems Engineer — Oct 2023 – Mar 2025 · Mumbai, India
Java, Spring Boot, Angular, Oracle SQL, message queues, REST APIs, JUnit, Tomcat

- Owned Margin Pledge features end to end for NSDL as one of three developers, from Angular screens through Spring Boot services to Oracle SQL queries and schema changes.
- Replaced per transaction type if else validation chains with a class based rule engine, rules for each pledge type defined in JSON config served from a config server, making a rule change a config change.
- Owned format validation across both entry paths: bulk files, checking presence, length, regex and identifier prefixes on every record and rejecting the file if any row failed, and the single pledge screen, validating in the front end before posting to the split file queue.
- Worked on the consolidation stage, holding the result until every parallel batch split from a file reports completion, then emitting the combined accepted and rejected output.
- Cut transaction latency 20% migrating a legacy Struts application to Angular and modular Spring Boot services, measured against the original in a test environment.
- Wrote JUnit suites across multiple in flight frameworks. Onboarded and mentored 4 engineers, authoring the onboarding documentation the team ramped up on.

PROJECTS

F1 Screenings Discovery Platform — github.com/dragonblade3k/f1-screenings
Next.js 14, TypeScript, Prisma, local LLM inference, SerpAPI, Claude Code

- Published 11 verified Formula 1 screenings from 70 candidate listings across 20 web sources, gating every event behind human review so nothing unverified reaches users.
- Traced malformed values in 73% of extracted rows to prompt based formatting, then made illegal output structurally impossible by constraining LLM decoding to a JSON schema.
- Made moderation an explicit state transition by modeling candidates and published events as separate entities rather than one table behind a boolean flag.

Content Pipeline — github.com/dragonblade3k/content-pipeline
Python, Ollama, Piper TTS, ffmpeg, Claude Code

- Automated a five stage pipeline from live F1 API to finished captioned vertical video through LLM scripting, neural narration, and ffmpeg assembly.
- Made data source, model, and voice each swappable through one environment variable, hosted or local, by placing every stage behind an interface.
- Built through an agentic coding loop, catching two failures that surfaced only against live services, including an undocumented API identifier mismatch; 28 tests, with regressions for both.

CancerNet, Histopathology Classification — github.com/dragonblade3k/Breast-Cancer-Classification
Python, TensorFlow, Keras, CUDA

- 86% accuracy and 0.91 ROC AUC over 99,906 held out patches, using a custom CNN of Conv/BatchNorm/ReLU blocks with depthwise separable convolutions and global average pooling.
- Added Grad CAM interpretability to surface the regions driving each prediction, making misclassifications diagnosable rather than opaque.

LEADERSHIP AND ACTIVITIES

Google Developer Student Club, RAIT · Community Engagement Head — Aug 2022 – May 2023
Grew event participation 30% running technical workshops and cross campus collaboration.

Athletics National level volleyball player, state level cricketer.

TECHNICAL SKILLS

Languages	Java, Python, C++, JavaScript, TypeScript, SQL
Fundamentals	Data structures, algorithms, complexity analysis, object oriented design, concurrency
Backend	Spring Boot, Node.js, REST API design, microservices, distributed services, Oracle SQL, MySQL, MongoDB, Prisma
Frontend	Next.js, React, Angular, HTML, CSS
AI tooling	Local and hosted LLM inference, schema constrained generation, LLM assisted development workflows
ML	TensorFlow, Keras, scikit-learn, NumPy, CUDA, OpenCV
Tools	Git, JUnit, Unix/Linux, Google Cloud Platform, Vercel